

## Smooth functions

recall definition and quadratic upper bound

$$f \text{ convex: } f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle$$

$$f \text{ } \beta\text{-smooth: } \|\nabla f(y) - \nabla f(x)\|_2 \leq \beta \cdot \|y - x\|_2$$

$$\Rightarrow f(y) \leq f(x) + \langle \nabla f(x), y - x \rangle + \frac{\beta}{2} \|y - x\|_2^2$$

---

## Smooth functions

gradient step guarantees improvement (Gradient Descent is a descent algorithm)

$$f(y) \leq f(x) + \langle \nabla f(x), y - x \rangle + \frac{\beta}{2} \|y - x\|_2^2$$

$$\text{GD} : x^+ = x - \eta \nabla f(x) \longrightarrow x^+ - x = -\eta \nabla f(x)$$

$$\begin{aligned} f(x^+) &\leq f(x) + \langle \nabla f(x), x^+ - x \rangle + \frac{\beta}{2} \|x^+ - x\|_2^2 \\ &= f(x) + -\eta \langle \nabla f(x), \nabla f(x) \rangle + \frac{\beta}{2} \cdot \eta^2 \|\nabla f(x)\|_2^2 \\ &= f(x) - \eta \|\nabla f(x)\|_2^2 + \frac{\beta}{2} \eta^2 \|\nabla f(x)\|_2^2 \\ &= f(x) - \underbrace{\eta \left(1 - \eta \cdot \frac{\beta}{2}\right)}_{\text{+ive (unless } x = x^*)}} \underbrace{\|\nabla f(x)\|_2^2}_{\text{+ive (unless } x = x^*)}} \end{aligned}$$

$$\text{If } \eta \text{ small enough: } \eta < \frac{2}{\beta} \text{ then } \eta \left(1 - \eta \cdot \frac{\beta}{2}\right) > 0$$

$$\begin{aligned} \text{Typically we choose } \eta = \frac{1}{\beta} : x^+ = x - \frac{1}{\beta} \nabla f(x) : \eta \left(1 - \eta \cdot \frac{\beta}{2}\right) &= \frac{1}{\beta} \left(1 - \frac{1}{2}\right) = \frac{1}{2\beta} \\ f(x^+) &\leq f(x) - \frac{1}{2\beta} \|\nabla f(x)\|_2^2 \end{aligned}$$

# Smooth functions

a bound on suboptimality of any point

If  $f$  is  $\beta$ -smooth:  $\frac{1}{2\beta} \|\nabla f(x)\|_2^2 \stackrel{(a)}{\leq} f(x) - f(x^*) \stackrel{(b)}{\leq} \frac{\beta}{2} \|x - x^*\|_2^2$

where  $x^*$  is solution to:  $\min: f(x)$

Again, use quad. upper bound:  $f(y) \leq f(x) + \langle \nabla f(x), y - x \rangle + \frac{\beta}{2} \|y - x\|_2^2$  \*

For (b): From \*:

$$f(x) \leq f(x^*) + \langle \nabla f(x^*), x - x^* \rangle + \frac{\beta}{2} \|x - x^*\|_2^2$$

$$\Rightarrow f(x) - f(x^*) \leq \frac{\beta}{2} \|x - x^*\|_2^2$$

For (a):  $f(x^*) \leq f(y) \leq \underbrace{f(x) + \langle \nabla f(x), y - x \rangle + \frac{\beta}{2} \|y - x\|_2^2}_{\text{minimize the quad. upper bound, over } y.}$

Find:  $\nabla f(x) + \beta(y - x) = 0$

$y = x - \frac{1}{\beta} \nabla f(x)$  . Now plug in this value of  $y$ :

$$\underline{f(x) - f(x^*) \geq \frac{1}{2\beta} \|\nabla f(x)\|_2^2}$$

## Smooth functions

co-coercivity: If  $f$  is  $\beta$ -smooth:  $\langle \nabla f(x) - \nabla f(y), x - y \rangle \geq \frac{1}{\beta} \|\nabla f(x) - \nabla f(y)\|_2^2$

To show this: (\*)  $f_x(z) = f(x) - \langle \nabla f(x), z \rangle$       $z^* = x$  is minimizer of  $f_x(z)$   
 $f_y(z) = f(y) - \langle \nabla f(y), z \rangle$       $z^* = y$  " "  $f_y(z)$

$$\begin{aligned}
 \textcircled{a} \bullet \underbrace{f(y) - (f(x) + \langle \nabla f(x), y - x \rangle)} &= \underbrace{f(y) - \langle \nabla f(x), y \rangle}_{f_x(y)} - \underbrace{(f(x) - \langle \nabla f(x), x \rangle)}_{f_x(x)} \\
 &= f_x(y) - f_x(x) \\
 &\geq \frac{1}{2\beta} \|\nabla f_x(y)\|_2^2 \\
 &= \underbrace{\frac{1}{2\beta} \|\nabla f(y) - \nabla f(x)\|_2^2}
 \end{aligned}$$

Adding  $\textcircled{a}$  and  $\textcircled{b}$   
 gives the  
 co-coercivity.

similarly, by flipping roles of  $x, y$ :

$$\textcircled{b} \bullet f(x) - (f(y) + \langle \nabla f(y), x - y \rangle) \geq \frac{1}{2\beta} \|\nabla f(y) - \nabla f(x)\|_2^2$$

## Strong convexity

recall definition and quadratic lower bound

$f$  is  $\alpha$ -strongly convex:  $\underbrace{f(x) - \frac{\alpha}{2} \|x\|_2^2}_{\text{is convex}}$

$$f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle + \frac{\alpha}{2} \|y - x\|_2^2$$

---

## Strong convexity

a bound on suboptimality of any point

$f$  is  $\alpha$ -strongly convex:

$$\frac{\alpha}{2} \|x - x^*\|^2 \stackrel{(a)}{\leq} f(x) - f(x^*) \stackrel{(b)}{\leq} \frac{1}{2\alpha} \|\nabla f(x)\|^2$$

For both (a) and (b) we use:  $f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle + \frac{\alpha}{2} \|y - x\|^2$

$$(a) \quad f(x) \geq f(x^*) + \langle \nabla f(x^*), x - x^* \rangle + \frac{\alpha}{2} \|x - x^*\|^2$$

$$f(x) - f(x^*) \geq \frac{\alpha}{2} \|x - x^*\|^2$$

$$(b) \quad f(x^*) \geq \min_y : f(x) + \langle \nabla f(x), y - x \rangle + \frac{\alpha}{2} \|y - x\|^2$$

Take grad., set to zero;

$$\nabla f(x) + \frac{\alpha}{2} (y - x) = 0$$

$$y = x - \frac{2}{\alpha} \nabla f(x) \rightarrow \text{plug back in}$$

$$f(x) - f(x^*) \leq \frac{1}{2\alpha} \|\nabla f(x)\|^2$$

## Strong convexity

coercivity

$f$  is  $\alpha$ -strongly convex:

$$\langle \nabla f(x) - \nabla f(y), x - y \rangle \geq \alpha \|x - y\|_2^2$$

Recall:  $f$  is  $\alpha$ -s.c. if  $g(x) = f(x) - \frac{\alpha}{2} \|x\|_2^2$  is convex

$g(x)$  convex  $\Leftrightarrow \nabla g(x)$  monotone

$$\langle \nabla g(x) - \nabla g(y), x - y \rangle \geq 0$$

$$\langle \nabla f(x) - \nabla f(y) - (\alpha x - \alpha y), x - y \rangle \geq 0$$

$$\Rightarrow \langle \nabla f(x) - \nabla f(y), x - y \rangle \geq \alpha \|x - y\|_2^2.$$



## Summary